MASENO 2025 KCSE PREDICTION

CONFIDENTIAL EXAM

233/3

CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

NAME…………………………………………………………………………………

SCHOOL………………………………………………… **SIGN**…………………………

INDEX NO…………………………………………… **ADM NO**……………………

(Kenya Certificate of Secondary Education)

INSTRUCTION TO THE CANDIDATES

a) Write your Name and Index Number, Admission Number and Class in the spaces provided at the top of this page.

b) Answer all the questions in the spaces in the spaces provided in this paper using English.

c) KNEC Mathematical tables and silent electronic calculators may be used.

d) All working MUST be clearly shown where necessary.

For Examiner's use only

Questions	Maximum score	Candidate's Score
1	22	
2	12	
3	06	
	40	

1. You are provided with:

- 4.0g of solid P, ethane-1,2-dioic acid with formula **$(COOH)_2.nH_2O$**.
- Solution Q, 0.2 M sodium hydroxide solution.

You are required to determine:

- The solubility of solid P.
- The relative formula mass of the acid **$(COOH)_2.nH_2O$**

Procedure I

(i) Fill the burette with distilled water.

(ii) Place all solid P into a boiling tube.

(iii) Transfer 4.0 cm^3 of distilled water from the burette into the boiling tube containing solid P.

(iv) Heat the mixture gently while stirring with a thermometer until all the solid dissolves.

(v) Allow the mixture to cool while stirring with the thermometer.

(vi) Record the temperature at which the crystals start to form in **table 1** below.

(vii) Add a further 2 cm^3 of distilled water from the burette into the boiling tube.

(viii) Repeat procedure (iv) to (vii) above and record the temperature crystals form to complete **table 1** below.

(Preserve the contents of the boiling tube for use in procedure II)

Table 1

Volume of distilled water (cm^3)	**Temperature at which crystals of solid P first appear. (^oC)**	**Solubility of solid P in g/100g of water**
4		
6		
8		
10		
12		

(6 marks)

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(a) On the grid below plot a graph of the solubility of solid P against temperature. **(3 marks)**

(b) From your graph determine:

(i) the solubility of solid P at 45^{o}C **(1 mark)**

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(ii) the temperature at which the solubility of solid P is 55g/100g of water. **(1 mark)**

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(c) How does the solubility of solid P vary with temperature? **(1 mark)**

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Procedure II

(i) Transfer all the contents of the boiling tube from **procedure I** into a clean 250ml volumetric flask. Rinse the boiling tube and transfer the rinsing into the volumetric flask.

(ii) Add distilled water to the mark.

(iii) Label the resulting solution as solution R.

(iv) Fill the burette with solution R.

(v) Pipette 25 cm^3 of solution Q into a clean conical flask. Add 3 drops of phenolphthalein indicator.

(vi) Titrate solution R against solution Q to an accurate end-point.

Record your results in **table 2** below.

Table 2

Titration	**I**	**II**	**III**
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of solution R used (cm^3)			

(4 marks)

(a) **Determine** the average volume of **solution R** used. **(1 mark)**

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(b) **Calculate**:

(i) The number of **moles** of solution Q used. **(1 mark)**

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(ii) Given that solution R is a dibasic acid, calculate the number of moles of **solution R** that reacted.

(1 mark)

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(iii) The **concentration** of **solution R** in moles per litre. **(1 mark)**

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(iv) Calculate the formula mass of acid **$(COOH)_2.nH_2O$,** hence find the value of n.

(H = 1, O=16, C = 12).

(2 marks)

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2. You **are provided** with **solid T,** which is a mixture of two compounds. **Carry out** the tests below. Write **your observations** and **inferences in the spaces** provided.

(a) Place about all **solid T** in a clean dry boiling tube. Add 5 ml of distilled water and shake thoroughly. Filter the mixture and keep both filtrate and residue. Divide the filtrate into three portions.

(i) Add aqueous NaOH dropwise to the first portion until in excess.

Observations	**Inference**
(1 mark)	(1 mark)

(ii) Add aqueous ammonia dropwise to the second portion until in excess.

Observations	**Inference**

(1 mark)	(1 mark)

(b) Rinse the residue with distilled water and transfer it into a clean boiling tube. Add the nitric (V) acid provided dropwise until all the solid dissolves.

Observations	**Inference**
(1 mark)	(1 mark)

(c) Divide the resulting solution into three portions.

(i) Add aqueous NaOH dropwise to the first portion until in excess.

Observations	**Inference**
(1 mark)	(1 mark)

(ii) Add aqueous ammonia dropwise to the second portion until in excess.

Observations	**Inference**
(1 mark)	(1 mark)

(iii) Add 3 drops of HCl to the third portion. Warm the mixture.

Observations	**Inference**

(1 mark)	(1 mark)

3. You are provided with **liquid W**. Carry out the following tests and record your observations and inferences in the spaces provided.

(a) Place five drops of **liquid W** on a clean dry watch glass and ignite it.

Observations	**Inference**
(1 mark)	(1 mark)

(b) Place $2cm^3$ of **liquid W** in a clean and dry test tube. Add all the sodium hydrogen carbonate provided.

Observations	**Inference**
(1 mark)	(1 mark)

(c) Place 2 cm^3 of **liquid W** in a clean and dry test tube. Add 1 cm^3 of acidified potassium dichromate (VI) and warm the mixture.

Observations	**Inference**

(1 mark)	(1 mark)

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